An Integrated Explainable AI Platform for Academic Guidance, Careers, and Wellbeing

R26-IT-073

Project Proposal Report

Component: AI-Based Student Mood Analysis, Stress Analyzer and Bio-Adaptive Study Recommendation

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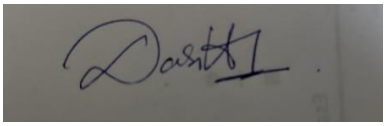
Sri Lanka Institute of Information Technology

Sri Lanka

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Many university students study to fixed timetables that ignore their physical and emotional condition on a given day. Sleep loss, elevated stress and low energy reduce the quality of study sessions, yet most study planners and consumer wearables treat these signals separately: wearables report stress and sleep scores, while planners schedule time without knowing the student's state. This creates a gap between what the body is signalling and how study time is planned.

This project proposes a Mood Analysis and Bio-Adaptive Study Recommendation component for the platform. It collects physiological readings from a smartwatch (heart rate, blood oxygen saturation, sleep duration, steps, calories burned, skin temperature, and stress level/score) through Firebase Firestore, and uses a Random Forest classifier to estimate the student's mood as Very Happy, Happy, Neutral, Stressed or Sad. A Cognitive Readiness Score is then calculated and mapped to four study modes (Peak Deep-Focus, Active Consolidation, Stress-Adaptive and Cognitive Recharge), each with a recommended study duration, break interval and activity type. Predictions are accompanied by simple explanations of which readings influenced the result.

The system will be developed with Python, Scikit-learn and FastAPI for the machine learning service, Firebase Firestore for data storage and synchronisation, and a React.js dashboard that presents biometric trends, study plans and a Pomodoro-style timer. The prototype is trained on a synthetic student wellbeing dataset, and the model is intentionally kept within a realistic accuracy range of about 70–80% so that reported performance is not inflated by the data-generation rules.

The expected outcomes are a working biometric-to-pedagogical pipeline that converts physiological data into practical study and recovery recommendations, supporting student self-regulation and helping to reduce cognitive overload. The system supports wellbeing and planning only; it is not a medical or diagnostic tool, and its real-world effect on fatigue or burnout would require separate evaluation.

Keywords: Mood Prediction, Wearable Biometrics, Random Forest, Cognitive Readiness, Adaptive Study Planning, Student Wellbeing.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form / Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CRS	Cognitive Readiness Score
CSV	Comma-Separated Values
DSR	Design Science Research
HR	Heart Rate
ML	Machine Learning
NPS	Net Promoter Score
PDPA	Personal Data Protection Act (Sri Lanka)
REST	Representational State Transfer
SpO_2	Peripheral Blood Oxygen Saturation
SUS	System Usability Scale
UI / UX	User Interface / User Experience

1. INTRODUCTION

1.1 Background and Context

University students in Sri Lanka balance heavy academic workloads, examinations, assignments and, increasingly, part-time work and internships. Study quality varies from day to day with sleep, stress and energy, yet most students plan their study time using fixed schedules or habits such as long unbroken revision blocks. When stress is high or sleep has been poor, long sessions can lead to cognitive overload, procrastination and fatigue, while on good days students may under-use their best focus periods.

Smartwatches now record heart rate, blood oxygen saturation, sleep, activity, skin temperature and stress indicators continuously. These readings are rich indicators of a person's physiological state, but they are normally shown as raw numbers or generic wellness scores. They are rarely converted into advice that a student can act on, such as how long to study, when to rest and what kind of task suits their current condition.

This component addresses that gap. It collects wearable readings through Firebase Firestore, predicts the student's mood with a Random Forest classifier, calculates a Cognitive Readiness Score, and recommends study durations, break intervals and activities through a React-based dashboard. It forms the wellbeing part of the wider platform and supports student self-regulation rather than replacing professional health advice.

The component aligns with SDG 3 (Good Health and Well-being) and SDG 4 (Quality Education), and fits within SLIIT's Intelligent Systems & Software Engineering research area through its use of machine learning, wearable data engineering and adaptive learning support.

1.2 Literature Review

Research in affective computing has long argued that physiological signals can reveal emotional and stress states [1]. Studies using wearable sensors and mobile phones have shown that stress and related states can be recognised from heart-rate based signals, skin responses, activity and phone usage [3], and public multimodal datasets such as WESAD have enabled systematic benchmarking of wearable stress and affect detection [4]. Related work on students has linked sleep, stress and behavioural patterns to academic performance and wellbeing, which motivates monitoring these factors in an academic setting.

On the modelling side, tree-based ensembles are widely used for tabular physiological data because they handle mixed feature types, resist overfitting better than single decision trees and

provide feature-importance measures that support explanation. The Random Forest algorithm [2], available in Scikit-learn [5], is therefore a suitable and interpretable choice for classifying mood from a small set of wearable features.

On the product side, consumer platforms such as Samsung Health, Fitbit and Garmin provide stress, sleep and readiness style scores, and time-management methods such as the Pomodoro technique [6] provide structured work and break cycles. However, these tools operate separately. Wearable apps do not schedule study sessions, and study timers apply fixed intervals regardless of the student's condition. Many studies of stress detection also concentrate on detection accuracy alone and stop short of translating the result into concrete behavioural recommendations or explaining it to the user.

1.3 Research Gap

Existing wearable and stress-detection research focuses on recognising a state, and existing study planners focus on scheduling time. What is missing is a unified, explainable pipeline that takes a student's physiological readings, estimates mood, calculates readiness for cognitive work and converts the result into specific study durations, break intervals and activity types, delivered through a student-friendly dashboard. There is also little work of this kind designed around the context of Sri Lankan university students.

This component fills that gap through a biometric-to-pedagogical translation pipeline: wearable data → mood prediction → Cognitive Readiness Score → adaptive study recommendation, with a short explanation of the readings that influenced each result.

2. OBJECTIVE

2.1 Main Objective

The main objective is to design, develop and evaluate an AI-based system that analyses students' wearable biometric data to predict mood, calculates a Cognitive Readiness Score, and recommends personalised study durations, breaks and activities, presented through an interactive analytics dashboard.

2.2 Specific Objectives

1. To build a biometric data pipeline that collects smartwatch readings through Firebase Firestore, normalises different data formats and handles missing or inconsistent fields, successfully processing at least 95% of incoming records in testing.
2. To train and evaluate a Random Forest classifier that predicts five mood classes (Very Happy, Happy, Neutral, Stressed, Sad), targeting a realistic test accuracy of about 70–80% and reporting a confusion matrix, per-class precision/recall and cross-validated results.
3. To design a Cognitive Readiness Score from physiological indicators and predicted mood, and map it to four study modes with defined study and break intervals.
4. To provide explanations for each prediction and recommendation by showing the most influential readings (using model feature importance) in plain language.
5. To develop a React.js dashboard, supported by a FastAPI service, that visualises biometric readings, stress trends, weekly study plans and progress, and integrates a Pomodoro-style timer, with API response times under 2 seconds in testing.
6. To evaluate usability with a controlled group of 20–30 students (with ethics approval and consent) using the System Usability Scale, targeting a mean SUS score of at least 70.

3. METHODOLOGY

3.1 Research Approach

The research follows the Design Science Research (DSR) methodology [7], [8] with its six steps: problem identification, objective definition, design and development, demonstration, evaluation and communication. Design Science is used to create the artefact (the mood analysis and recommendation system), and experimental evaluation is used to test the machine learning model and the recommendation logic.

3.2 Technical Methods and Algorithms

- **Data collection and preprocessing:** Smartwatch telemetry is stored in Firebase Firestore. Pandas and NumPy are used to normalise formats and units, impute missing values (median imputation), remove clearly invalid readings and encode categorical labels.
- **Mood classification:** A Random Forest classifier [2] trained with Scikit-learn [5] on the physiological features (heart rate, SpO_2, sleep duration, steps, calories burned, skin temperature, and stress level/score). Data is split with stratification into training and test sets, and hyperparameters are tuned with cross-validation. The trained model and encoders are saved with Joblib.
- **Cognitive Readiness Score (CRS):** A weighted score (0–100) built from normalised sleep, heart rate, SpO_2, stress indicators and the predicted mood. The initial weights are design choices to be reviewed with the supervisor and refined during testing; the score is a planning aid, not a clinical measure.
- **Adaptive study modes:** The CRS is mapped to one of four modes (Table 1). The band boundaries are provisional and will be tuned during evaluation.
- **Explainability:** Global feature importance from the Random Forest and a per-reading summary of the most influential indicators, shown on the dashboard in plain language.
- **API and dashboard:** The model is exposed through FastAPI REST endpoints and consumed by a React.js dashboard.

Study mode	Provisional CRS band	Study duration	Break duration	Recommended approach
Peak Deep-Focus	75–100	50 min	10 min	Complex problem-solving
Active Consolidation	55–74	40 min	5 min	Concept revision and exercises
Stress-Adaptive	35–54	25 min	10 min	Light revision and flashcards

Study mode	Provisional CRS band	Study duration	Break duration	Recommended approach
Cognitive Recharge	0–34	15 min	15 min	Light learning and recovery

Table 1: Adaptive Study Modes

3.3 Tools and Platforms with Justification

Technology	Purpose	Justification
Python	ML and data processing	Mature ecosystem for data science and machine learning
Scikit-learn	Model training, encoding, evaluation	Reliable implementations of Random Forest, cross-validation and metrics
Pandas and NumPy	Preprocessing and numerical operations	Efficient handling of tabular sensor data
Joblib	Saving and loading model artefacts	Simple, fast persistence of trained models and encoders
FastAPI	REST API for the ML model	High-performance, easy to document and integrate with the dashboard
Firebase Firestore	Storage and synchronisation of biometric records	Real-time sync, flexible schema for sensor data and a free tier suitable for a prototype
React.js	Analytics dashboard	Component-based UI suited to charts, trends and interactive planning
Samsung Galaxy Watch	Source of physiological telemetry	Provides heart rate, SpO_2, steps, calories, temperature, sleep and stress readings

Table 2: Technology Justification Table

3.4 Data Collection and Ethical Considerations

- The prototype is trained on a synthetic student wellbeing dataset (student_wellbeing_dataset.csv) generated to reflect plausible ranges of the physiological features and mood classes.
- Because the labels in a synthetic dataset follow the generation rules, accuracy on this data does not prove real-world performance. The model is deliberately kept within a realistic range rather than tuned to maximise accuracy, and this limitation will be stated in the report.
- Any collection of real wearable data or student feedback will take place only after approval from the SLIIT Research Ethics Committee, with informed consent, anonymisation and the right to withdraw.
- Physiological data is sensitive. Data handling will follow Sri Lanka's Personal Data Protection Act [10], with minimal retention, access controls and no sharing outside the platform.

- The system gives wellbeing and study-planning suggestions only. It does not diagnose conditions, and students experiencing persistent distress will be advised to seek support from qualified professionals.

3.5 Validation Strategy and Performance Metrics

- **Model performance:** Test accuracy of about 70–80%, with macro-averaged F1-score, confusion matrix and 5-fold cross-validation results.
- **Pipeline robustness:** At least 95% of test records (including records with missing or inconsistent fields) processed without failure.
- **Recommendation consistency:** Unit tests confirming that each CRS band returns the correct study mode and intervals, plus review of sample scenarios with the supervisor.
- **Performance:** API response time under 2 seconds and dashboard load time under 3 seconds on a standard laptop.
- **Usability:** SUS [9] with 20–30 students and additional questions on clarity of explanations; target mean SUS of at least 70.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

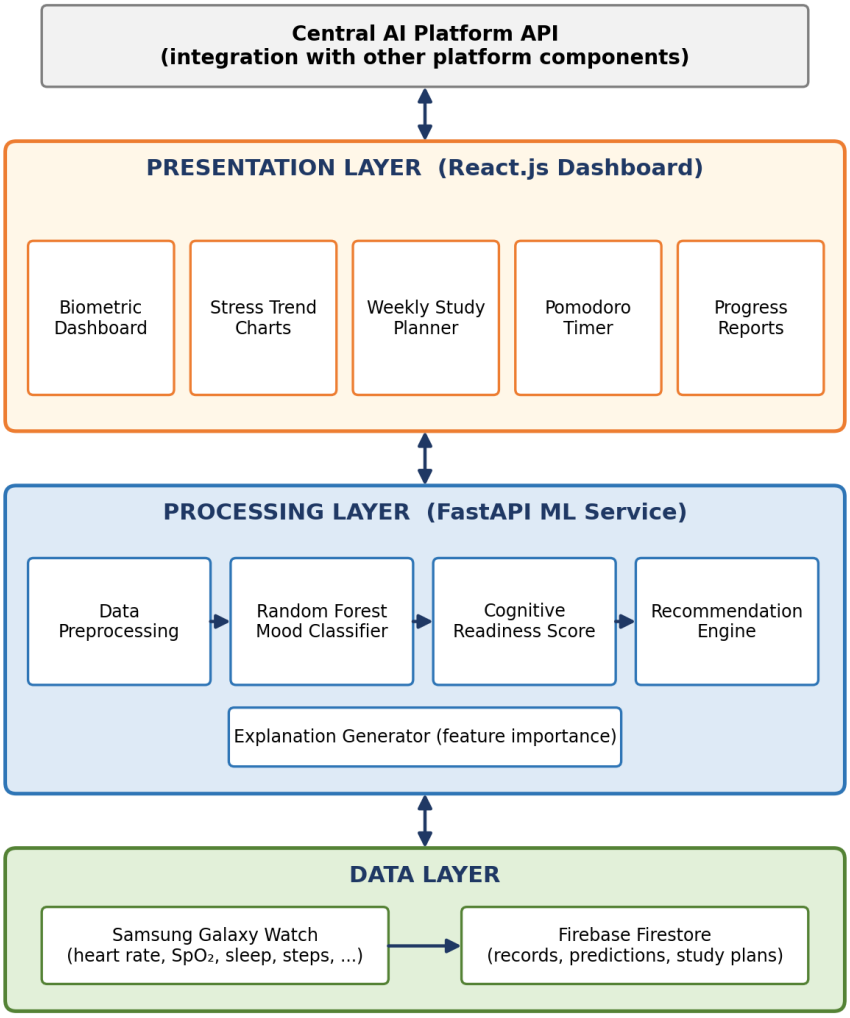


Figure 1: System Architecture Diagram

4.1 Individual Component and Their Parts

- **Data Layer:** Smartwatch telemetry and Firebase Firestore storing biometric records, predicted moods and study plans.
- **Processing Layer (FastAPI):** Preprocessing module, Random Forest mood classifier, Cognitive Readiness Score calculator, and study recommendation engine with explanation generator.
- **Presentation Layer (React.js):** Biometric dashboard, stress trend charts, weekly study planner, progress reports and Pomodoro timer.
- **Integration Layer:** Endpoints that share mood and readiness results with the rest of the platform.

4.2 Component Interactions

- The dashboard requests the latest readings, mood and recommendation from the FastAPI service.
- The FastAPI service reads biometric records from Firestore, runs preprocessing and prediction, and returns the mood, CRS, study mode and explanation.
- Recommendations are saved back to Firestore so that history and trends can be displayed.
- The Pomodoro timer uses the recommended study and break durations for the current session.

4.3 Data Flow Explanation

1. The smartwatch records physiological readings and syncs them to Firebase Firestore.
2. The preprocessing module normalises formats and handles missing or inconsistent fields.
3. The Random Forest model predicts the mood class from the processed features.
4. The CRS is calculated and mapped to a study mode.
5. The recommendation engine returns study duration, break interval, activity type and a short explanation.
6. The React dashboard displays results, trends and the study plan, and records progress.

4.4 Scalability and Security Considerations

- **Scalability:** Firestore scales automatically for concurrent users, and the stateless FastAPI service can be containerised and scaled horizontally.
- **Security:** Authenticated access to records, per-user data isolation using Firestore security rules, encrypted transport (HTTPS), and minimal retention of sensitive readings in line with the PDPA [10].

5. USER REQUIREMENTS

5.1 Stakeholder Identification

- **Primary stakeholders:** University students who use the dashboard and recommendations.
- **Secondary stakeholders:** Supervisors and university administration (oversight and integration), student counsellors (potential future users of aggregated, anonymised trends), and system administrators.

5.2 Functional Requirements

1. The system shall collect biometric readings from the smartwatch and Firebase Firestore.
2. The system shall normalise data formats and handle missing or inconsistent fields.
3. The system shall predict the student's mood as Very Happy, Happy, Neutral, Stressed or Sad.
4. The system shall calculate a Cognitive Readiness Score and assign one of four study modes.
5. The system shall recommend study duration, break interval and activity type for the assigned mode.
6. The system shall explain which readings most influenced the result.
7. The system shall display biometric readings, historical stress trends, study activity and progress reports on a dashboard.
8. The system shall provide a weekly study planner and a Pomodoro timer driven by the recommendations.
9. The system shall expose its results through REST APIs for integration with the wider platform.

5.3 Non-Functional Requirements

- **Performance:** API responses within 2 seconds; dashboard load within 3 seconds.
- **Security and privacy:** Authenticated access, per-user data isolation, encrypted transport, and compliance with the PDPA.
- **Usability:** Clear, non-technical language, with a target mean SUS score of at least 70.
- **Reliability:** Graceful handling of missing sensor data, with sensible defaults and clear messages.
- **Transparency and safety:** Recommendations are presented as suggestions, with a visible note that the system is not a medical tool.

- **Compatibility:** Dashboard works on current versions of Chrome, Edge and Firefox, and is responsive on mobile browsers.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary market is Sri Lankan undergraduates aged 18–25, especially in IT, engineering and business programmes at universities such as SLIIT, Colombo, Moratuwa and private campuses. Secondary markets include universities' student support units and wellness programmes.

Attribute	Description
Name	Nimal Perera (sample)
Age	21–23
Education	Second- or third-year BSc IT undergraduate
Location	Urban or semi-urban Sri Lanka (e.g., Colombo, Gampaha, Negombo)
Pain points	Irregular sleep, exam stress, long unproductive study sessions, no guidance on when to rest
Goals	Study more effectively, manage stress, balance study and recovery
Tech behaviour	Owns a smartphone and smartwatch or fitness band, uses WhatsApp and Instagram, prefers simple dashboards
Willingness to pay	Free basic use; low monthly fee for premium analytics

Table 3: Sample Persona

6.2 Value Proposition

The system turns wearable data a student already has into personalised, explainable study advice: when to study hard, when to slow down and when to recover.

6.3 Revenue Model

Freemium model (indicative figures, to be validated):

- **Free:** Basic mood analysis and daily recommendation.
- **Premium (about LKR 750/month):** Long-term trend analytics, weekly reports and advanced planner features.
- **Partnerships:** University licensing for student wellbeing programmes, using anonymised aggregate insights with consent.

6.4 Cost Estimation

- Development and maintenance: LKR 200,000
- Cloud (Firebase and hosting): LKR 100,000/year
- Marketing: LKR 100,000/year

- Legal/IP and data-protection review: LKR 75,000
- **Total first-year estimate: LKR 475,000**

6.5 Pricing Strategy

Freemium entry to build the user base, student-friendly monthly pricing with annual discounts, and discounted licences through university partnerships.

6.6 Competitive Advantage

The pipeline links wearable data directly to study planning, gives explanations rather than bare scores, integrates with a Pomodoro-style planner, and is designed around the needs of Sri Lankan students.

6.7 Intellectual Property Considerations

Trademark the application name, copyright the source code, and use appropriate open-source licences for libraries such as React, FastAPI and Scikit-learn.

7. BUDGET AND JUSTIFICATION

Item	Estimated Cost (LKR)	Subtotal (LKR)
Hardware Costs	0 – 120,000	0 – 120,000
Software Licenses	0	0
Cloud Services	0 – 20,000	0 – 20,000
Human Effort Cost	0	0
Miscellaneous	15,000 – 35,000	15,000 – 35,000
Total Estimated Budget		15,000 – 175,000

Table 4: Budget and Justification Table

Hardware (LKR 0 – 120,000): A smartwatch that reports heart rate, SpO_2, skin temperature, sleep and stress is required for collecting real readings; the cost is zero if an existing device is used. A laptop is assumed to be available.

Software licences (LKR 0): Python, Scikit-learn, Pandas, NumPy, FastAPI, React.js, VS Code and Google Colab are free or open-source.

Cloud services (LKR 0 – 20,000): Firebase Firestore free tier is expected to cover prototype storage and synchronisation; the upper range covers optional hosting or a domain for demonstration.

Human effort (LKR 0): Development is carried out by the student as part of the final-year project.

Miscellaneous (LKR 15,000 – 35,000): Printing and binding of proposal and report documents, poster and exhibition materials, and small incentives for volunteers in usability testing.

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1 Task Breakdown

Task ID	Task Description	Main Tasks	Responsible	Estimated Effort
1.0	Project Setup	Literature review, requirement gathering, environment and Firebase setup	Withanage D.K.	3 weeks
2.0	Data Preparation	Synthetic dataset design, cleaning, feature selection	Withanage D.K.	3–4 weeks
3.0	Mood Prediction Model	Random Forest training, tuning, evaluation, model saving	Withanage D.K.	5 weeks
4.0	Readiness and Study Modes	CRS design, study mode mapping, recommendation logic	Withanage D.K.	4 weeks
5.0	Backend Development	FastAPI endpoints, Firestore integration	Withanage D.K.	5–6 weeks
6.0	Dashboard Frontend	React dashboard, charts, planner, Pomodoro timer	Withanage D.K.	7–8 weeks
7.0	Explainability and Integration	Feature-importance explanations, platform API integration	Withanage D.K.	3 weeks
8.0	Testing and Validation	Model evaluation, unit and API tests, usability testing (SUS)	Withanage D.K.	5 weeks
9.0	Documentation and Demo	Final report, poster, demo video, exhibition preparation	Withanage D.K.	4 weeks

Table 5: Work Breakdown Structure

8.2 Timeline Alignment

Month	Key Tasks	Milestone
Months 1–2	Tasks 1.0 + 2.0	Environment ready, dataset prepared
Months 2–4	Tasks 3.0 + 5.0	Model trained, backend working
Months 4–5	Task 4.0	Readiness score and study modes complete
Months 4–6	Task 6.0	Dashboard working
Months 6–7	Task 7.0 + Task 8.0 (first round)	Integrated prototype
Months 7–8	Task 8.0 (full testing) + Task 9.0	Evaluation complete, final report and exhibition ready

Table 6: Timeline Alignment Table

8.3 Balanced Workload Distribution

This component is carried out by one student, who handles all tasks. Core features (data pipeline, model and recommendation logic) are completed first, followed by the dashboard, and finally testing and documentation. Buffer time is included in each phase for revisions, and weekly effort is planned at 10–15 hours around other study commitments.

9. GANTT CHART

Task	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct
1.0 Project Setup								
2.0 Data Preparation								
3.0 Mood Prediction Model								
4.0 Readiness and Study Modes								
5.0 Backend Development								
6.0 Dashboard Frontend								
7.0 Explainability and Integration								
8.0 Testing and Validation								
9.0 Documentation and Demo								

Figure 2: Gantt Chart

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11. APPENDICES

Appendix A: Survey Instruments – Usability Testing Questionnaire (System Usability Scale – SUS)

To be used after demonstrating the dashboard to 20–30 students (1 = Strongly Disagree, 5 = Strongly Agree).

1. I think that I would like to use this system frequently.
2. I found the system unnecessarily complex.
3. I thought the system was easy to use.
4. I think that I would need the support of a technical person to be able to use this system.
5. I found the various functions in this system were well integrated.
6. I thought there was too much inconsistency in this system.
7. I would imagine that most people would learn to use this system very quickly.
8. I found the system very cumbersome to use.
9. I felt very confident using the system.
10. I needed to learn a lot of things before I could get going with this system.

Additional Questions

- How clear were the explanations of why a study mode was recommended? (1–5 + comment)
- Were the recommended study and break durations realistic for you? (1–5 + comment)
- Did you feel comfortable with how your biometric data was used and protected? (1–5 + comment)
- Any suggestions for improving the dashboard or study planner?

Appendix B: Detailed Diagrams

1. Data Flow Diagram

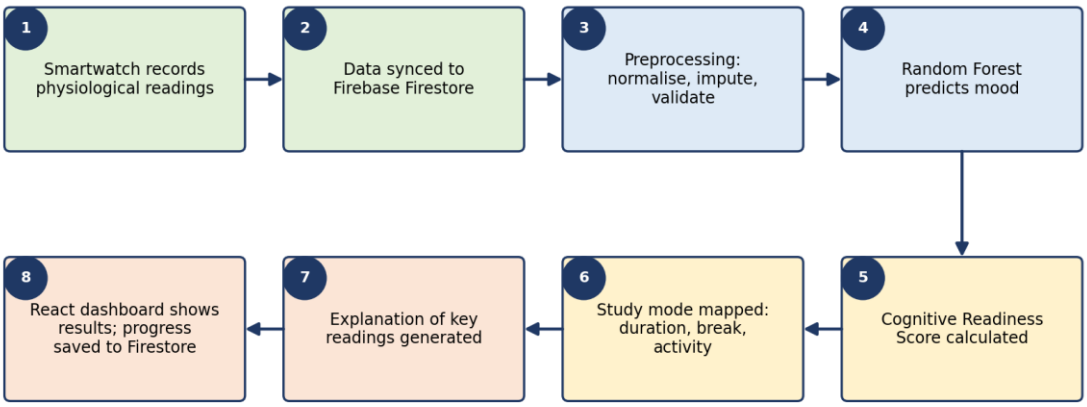


Figure 3: Data Flow Diagram

2. Entity Relationship Diagram

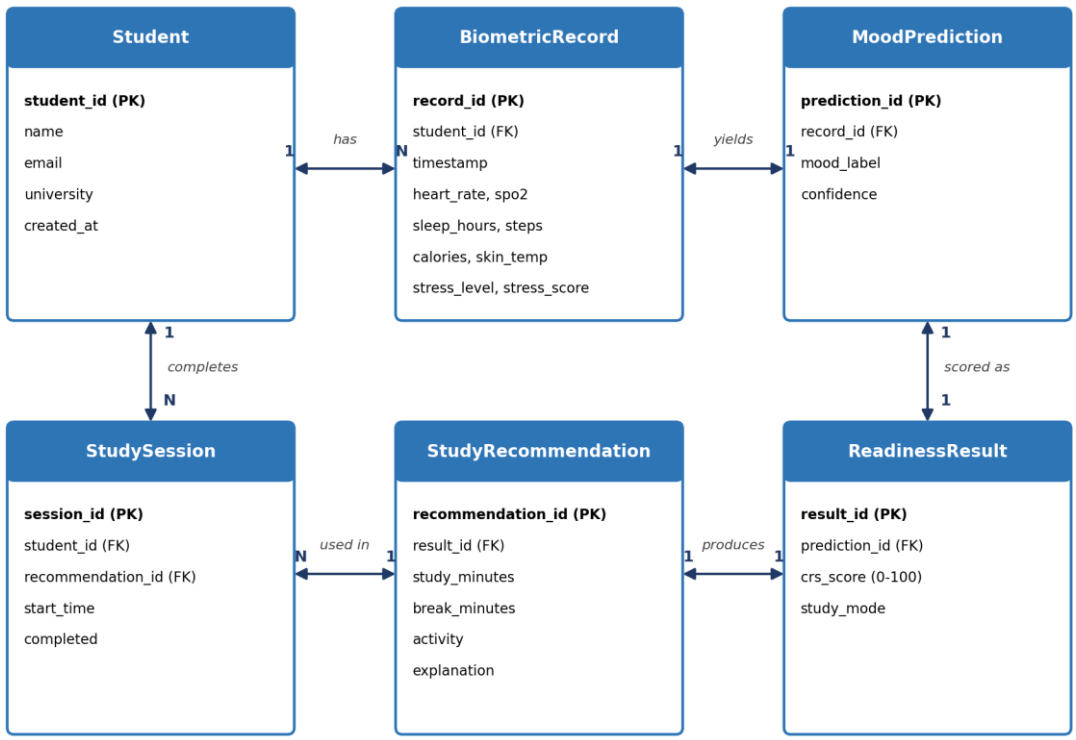


Figure 4: Entity Relationship Diagram

Appendix C: Preliminary UI Mockups

The mockups below show the planned React.js web dashboard screens. Values shown are sample data for illustration only.

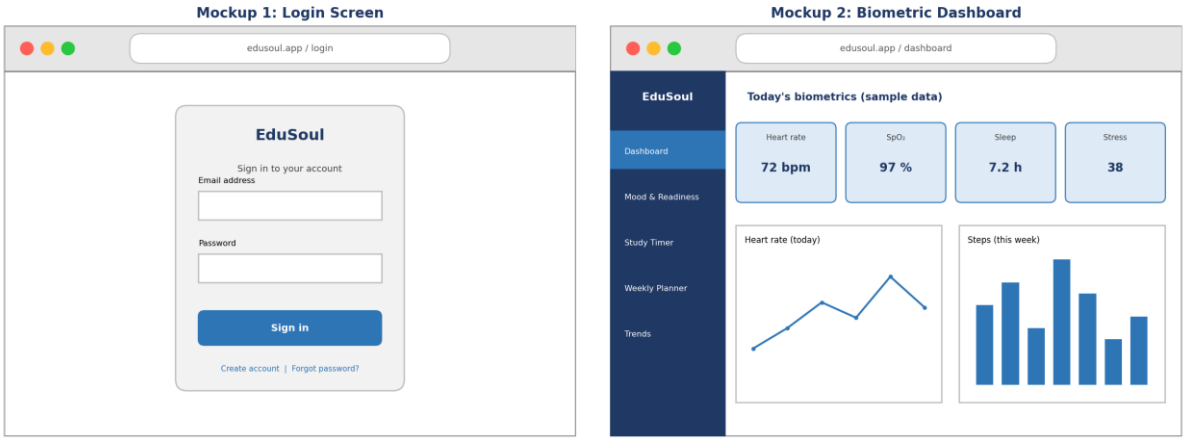


Figure 5.1: Preliminary UI Mockups (Mockups 1–2)

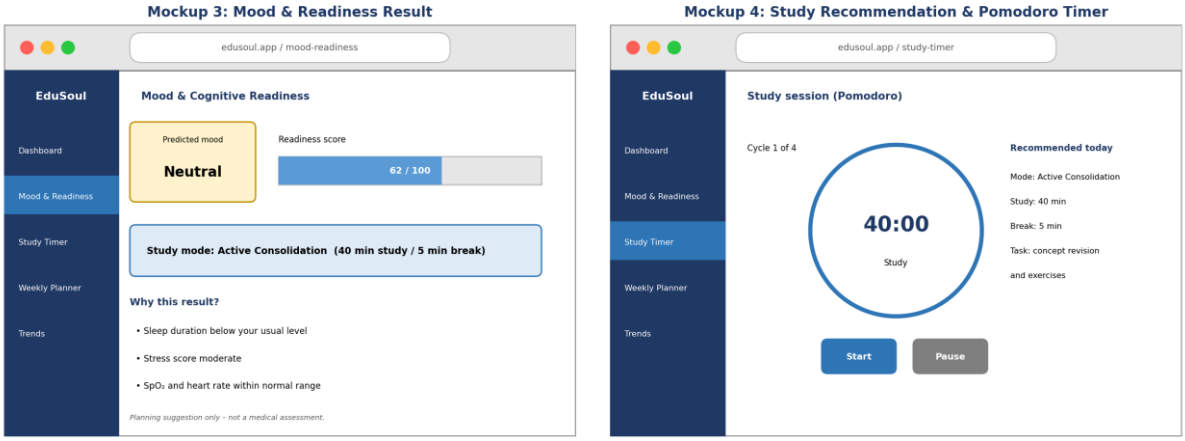


Figure 5.2: Preliminary UI Mockups (Mockups 3–4)

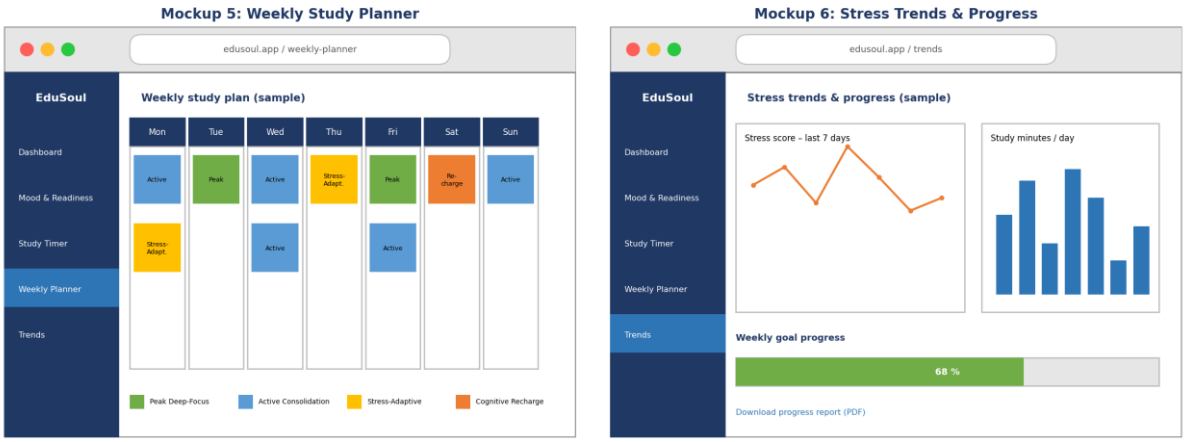


Figure 5.3: Preliminary UI Mockups (Mockups 5–6)

Appendix D: Risk Analysis

Risk ID	Risk Description	Probability	Impact	Mitigation Plan
R1	Model accuracy on synthetic data does not reflect real-world performance	High	High	State the limitation clearly, keep the model within a realistic accuracy range, and validate with consented real data if ethics approval is obtained
R2	Wearable data is missing, noisy or inconsistent	Medium	High	Validation rules, imputation and clear user messages in the preprocessing pipeline
R3	Low number of volunteers for usability testing	Medium	Medium	Recruit early through university groups and offer small incentives
R4	Privacy concerns about physiological data	Medium	Very High	Ethics approval, informed consent, minimal retention, per-user access rules
R5	Recommendations misunderstood as medical advice	Medium	High	Visible disclaimer, supportive wording, and guidance to seek professional help when needed
R6	Delay due to other academic commitments	Medium	Medium	Two-week buffer in the timeline; prioritise core pipeline first

Table 7: Risk Analysis Table

Appendix E: Ethical Documents

Ethics Clearance Plan

- Submit an ethics application to the SLIIT Research Ethics Committee before any user testing or real data collection.
- All participation will use informed consent.
- Prototype development uses synthetic data only; no real personal data is collected before approval.
- Participants can withdraw at any time, and all feedback and readings will be anonymised.
- The system will not be presented as a diagnostic or medical device.